Check for updates

Germline and Somatic Genomic Testing for Metastatic Prostate Cancer: ASCO Guideline

Evan Y. Yu, MD[1]; R. Bryan Rumble, MSc[2]; Neeraj Agarwal, MD[3]; Heather H. Cheng, MD, PhD[1]; Scott E. Eggener, MD[4]; Rhonda L. Bitting, MD[5]; Himisha Beltran, MD[6]; Veda N. Giri, MD[7]; Daniel Spratt, MD[8]; Brandon Mahal, MD[9]; Kevin Lu, MD[10]; Tony Crispino[11]; and Edouard J. Trabulsi, MD[12]

DOI https://doi.org/10.1200/JCO-24-02608

ABSTRACT

ASCO Guidelines provide recommendations with comprehensive review and analyses of the relevant literature for each recommendation, following the guideline development process as outlined in the ASCO Guidelines Methodology Manual. ASCO Guidelines follow the ASCO Conflict of Interest Policy for Clinical Practice Guidelines.

Clinical Practice Guidelines and other guidance ("Guidance") provided by ASCO is not a comprehensive or definitive guide to treatment options. It is intended for voluntary use by clinicians and should be used in conjunction with independent professional judgment. Guidance may not be applicable to all patients, interventions, diseases or stages of diseases. Guidance is based on review and analysis of relevant literature, and is not intended as a statement of the standard of care. ASCO does not endorse third-party drugs, devices, services, or therapies and assumes no responsibility for any harm arising from or related to the use of this information. See complete disclaimer in Appendix 1 and 2 (online only) for more.

PURPOSE To evaluate evidence on germline and somatic genomic testing for patients with metastatic prostate cancer and provide recommendations.

METHODS A systematic review by a multidisciplinary panel with patient representation was conducted. The PubMed database was searched from January 2018 to May 2024. Articles were selected for inclusion if they reported on patients with metastatic prostate cancer who received a germline or somatic genomic test and/or made comparisons between those tests, reported detection rates, prognostic information, or treatment implications.

RESULTS A total of 1,713 papers were identified in the literature search. After applying the eligibility criteria, 14 remained: eight systematic reviews and six clinical trials.

RECOMMENDATIONS Patients with metastatic prostate cancer should undergo both germline and somatic DNA sequencing using panel-based assays. These tests can guide the use of poly(ADP-ribose) polymerase inhibitors, which have a survival benefit in metastatic castration-resistant prostate cancer. In addition, germline testing may have screening implications for additional cancers for patients and cascade testing implications for family members. The data supporting when to perform repeat testing and optimal tissue type to use (eg, primary tumor *v* metastatic biopsy versus circulating tumor DNA [ctDNA] testing) are more limited, but this panel recommends considering retesting in patients whose results were previously negative or uninformative, and to consider either a metastatic biopsy or ctDNA when a significant change in clinical status occurs. Next-generation genomic sequencing findings that are associated with prognostic only (and not predictive) value should not be used to guide treatment outside of a clinical trial.

Additional information is available at www.asco.org/genitourinary-cancer-guidelines.

ACCOMPANYING CONTENT

Listen to the podcast by Brittany Harvey and Dr Yu at https://ascopubs.org/do/germline-and-somatic-genomic-testing-metastatic-prostate-cancer-guideline

Appendix

Data Supplement

Accepted December 3, 2024
Published January 9, 2025

Evidence Based Medicine Committee approval: November 25, 2024

J Clin Oncol 43:748-758

© 2025 by American Society of Clinical Oncology

View Online Article

Downloaded from ascopubs.org by 70.187.168.80 on February 2, 2026 from 070.187.168.080
Copyright © 2026 American Society of Clinical Oncology. All rights reserved.

TARGET POPULATION AND AUDIENCE

Target Population
Patients with metastatic prostate cancer.

Target Audience
Medical oncologists, radiation oncologists, urologists, pathologists, geneticists and genetic counselors, nurses, other health care practitioners, social workers, patients, and caregivers.

INTRODUCTION

Prostate cancer is a major cause of disease burden, with an estimated approximately 299,000 new cases (the highest incidence in males, comprising 29% of all new cases) and approximately 35,000 deaths (the second highest cause of mortality in males and constituting 11% of all cancer-related deaths) in 2024.[1] As with many other cancers, somatic genomic testing has both prognostic and predictive capability and may affect treatment decision making for patients with metastatic prostate cancer. Germline genetic testing may also provide similar information, with the added potential to offer additional hereditary cancer risk information with relevance to family members regarding cancer risk beyond prostate (eg, breast, pancreas, colon, endometrial, etc). For this reason, efforts to identify germline and somatic genomic tests that provide the most useful information for specific patient situations have been investigated.

This guideline addresses the following questions: why germline and somatic genomic testing should be offered, the criteria for which patients should be offered testing, which test(s) to use, when in the disease course testing should be considered, and which biospecimens should be used for testing.

GUIDELINE QUESTIONS

This clinical practice guideline addresses six overarching clinical questions: (1) Who should receive germline testing with next-generation sequencing (NGS) technologies? (2) Who should receive somatic testing with NGS technologies? (3) Who should receive sequential somatic testing with NGS technologies? (4) What are the strengths and weaknesses of primary tumor archival tissue versus fresh metastatic biopsy tissue versus circulating tumor DNA (ctDNA) testing for somatic testing? (5) What are the key therapeutic implications of germline or somatic testing for single-gene genetic variants (eg, *BRCA1*, *BRCA2*)? (6) What are the key prognostic implications of germline or somatic testing?

METHODS

Guideline Development Process

This systematic review-based guideline product was developed by a multidisciplinary Expert Panel, which included a patient representative and an ASCO guidelines staff member with health research methodology expertise (Appendix Table A1, online only).

The recommendations were developed by using a systematic review of evidence identified through online searches of PubMed (initial searches completed June 2023, from Jan 2018 through to end of May 2023; final searches ran June 2024) and clinical experience. Articles were selected for inclusion in the systematic review based on the following criteria:

- Population: Patients with metastatic prostate cancer.
- Interventions: Germline and somatic genomic tests.
- Comparisons: Studies examining germline tests, studies examining somatic genomic tests, and studies where both germline and somatic tests were performed.
- Outcomes: Detection rates, incidence rates, prognostic information, subsequent treatment.
- Sample size: Minimum 25 per arm desired.

Articles were excluded from the systematic review if they were (1) meeting abstracts not subsequently published in peer-reviewed journals; (2) editorials, commentaries, letters, news articles, case reports, narrative reviews; and (3) published in a non-English language.

Twelve full-panel meetings were held, and members were asked to provide ongoing input on the guideline development protocol, quality and assessment of the evidence, generation of recommendations, draft content, as well as review and approve drafts during the entire development of the guideline. ASCO staff met routinely with the Expert Panel co-chairs and corresponded with the panel via e-mail to coordinate the process to completion. Ratings for strength of the recommendation and evidence quality are provided with each recommendation, defined in Appendix Table A2. The quality of the evidence for each outcome was assessed using the Cochrane Risk of Bias tool and elements of the GRADE quality assessment and recommendations development process.[2,3] GRADE quality assessment labels (ie, high, moderate, low, and very low) were assigned for each outcome by the project methodologist in collaboration with the Expert Panel co-chairs and reviewed by the full Expert Panel. All funding for the administration of the project was provided by ASCO.

Guideline Review and Approval

The draft recommendations were released to the public for open comment from August 23, 2024, through September 6, 2024. Response categories of "Agree as written," "Agree with suggested modifications," and "Disagree. See comments"

Downloaded from ascopubs.org by 70.187.168.80 on February 2, 2026 from 070.187.168.080
Copyright © 2026 American Society of Clinical Oncology. All rights reserved.

were captured for every proposed recommendation along with 17 written comments received. A total of 100% of the five respondents either agreed or agreed with slight modifications to the recommendations and none of the respondents disagreed. Expert Panel members reviewed comments from all sources and determined whether to maintain original draft recommendations, revise with minor language changes, or consider major recommendation revisions.

A guideline implementability review was conducted. Based on this review, revisions were made to the draft to clarify recommended actions for clinical practice.

All changes were incorporated into the final manuscript prior to ASCO Evidence Based Medicine Committee (EBMC) review and approval. All ASCO guidelines are ultimately reviewed and approved by the Expert Panel and the ASCO EBMC before submission to the Journal of Clinical Oncology for editorial review and consideration for publication.

Guideline Updating

The ASCO Expert Panel and guidelines staff will work with co-chairs to keep abreast of any substantive updates to the guideline. Based on formal review of the emerging literature, ASCO will determine the need to update. The ASCO Guidelines Methodology Manual (available at www.asco.org/guideline-methodology) provides additional information about the guideline update process. This is the most recent information as of the publication date.

RESULTS

From the trials obtained in this search, the administration of a poly(ADP-ribose) polymerase (PARP) inhibitor such as olaparib, talazoparib, niraparib, or rucaparib demonstrated improvement in radiographic progression-free survival in patients with metastatic prostate cancer with homologous recombination deficiency (HRD; Data Supplement, Table S4, online only). For complete results, Key Outcomes of Interest are provided in the Data Supplement. The PROfound trial demonstrated an overall survival benefit of olaparib compared to androgen receptor pathway inhibitor (ARPI) switch to abiraterone or enzalutamide for patients with metastatic castration-resistant prostate cancer (mCRPC) with HRD.[4]

Characteristics of Studies Identified in the Literature Search

A total of 1,713 papers were identified in the literature search. After applying the eligibility criteria, 14[4-17] remained, comprising eight[5-12] systematic reviews and six trials.[4,13-17] Those 14 papers form the evidentiary basis for the guideline recommendations.

The identified trials were published between 2020 and 2024. The randomized trials compared similar interventions. The primary outcome for all the trials was progression-free survival as assessed by radiographic or image-based means. Characteristics of the included studies are in the Data Supplement (Table S1a): Characteristics of Studies Identified in the Search of the Literature: Systematic Reviews, and the Data Supplement (Table S1b): Characteristics of Studies Identified in the Search of the Literature: Trials.

Evidence Quality Assessment

The quality of evidence was assessed for each outcome of interest. This rating includes factors such as study design, consistency of results, directness of evidence, precision, publication bias, and magnitude of effect, assessed by one reviewer. Evidence quality ratings for the outcomes of interest are provided in the Data Supplement (Table S3): Study Quality Assessment. Refer to Appendix Table A2 for definitions for the quality of the evidence, and the Methodology Manual for more information.

Of the six trials,[4,13-17] four (TALAPRO-2,[13] MAGNITUDE,[14] PROpel,[15] and PROfound[4]) were deemed to have a high certainty level and two (BRCAAWAY and TRITON-3[16]) a moderate level of certainty. Rationale for the moderate determination in the two trials was around small sample size (BRCAAWAY) and blinding and/or concealment (BRCAAWAY and TRITON3).

Evidence Summary

The evidence base that informed the recommendations comprised eight[5-12] systematic reviews and six trials.[4,13-17] When looking at the evidence obtained in totality, the trials showed that the administration of a PARP inhibitor (such as olaparib, talazoparib, niraparib, or rucaparib) results in increases in progression-free survival in patients with mCRPC as measured by radiographic or other imaging means. The most compelling evidence was found in the PROFOUND trial, which demonstrated an overall survival benefit for patients with mCRPC with HRD, for the PARP inhibitor, olaparib, when compared to ARPI switch to abiraterone or enzalutamide.[4] A consistent observation across these studies is that the *BRCA1/2* alteration subsets were most predictive of favorable PARP inhibitor outcomes.

RECOMMENDATIONS

All recommendations are available in Table 1.

General Considerations

The recommendations provided in this guideline are predicated on the following:

- The patient has a life expectancy of more than 6 months
- The patient is a candidate for systemic treatment
- Appropriate germline and somatic testing, including expertise in interpretation, is readily available.

Downloaded from ascopubs.org by 70.187.168.80 on February 2, 2026 from 070.187.168.080
Copyright © 2026 American Society of Clinical Oncology. All rights reserved.

TABLE 1. Summary of All Recommendations

Clinical Question	Recommendation
General note. The following recommendations (strong or conditional/weak) and terminology (see Data Supplement) represent reasonable options for patients depending on clinical circumstances and in the context of individual patient preferences. Recommended care should be accessible to patients whenever possible	
Who should receive germline testing with NGS technologies?	1. All patients with metastatic prostate cancer should undergo germline genetic testing with next-generation sequencing technologies. (Evidence quality: High; Strength of recommendation: Strong)
Who should receive somatic testing with NGS technologies?	2. Those patients with metastatic prostate cancer (both CSPC and CRPC) who are being considered for biomarker-directed systemic treatment should undergo somatic testing with next-generation sequencing technologies. (Evidence quality: High; Strength of recommendation: Strong)
	Practical information for Recommendation 2: While there are no current FDA-approved biomarker-directed treatments following somatic testing for mCSPC, somatic testing may be warranted in the presence of high-volume disease or where there is a high likelihood the patient's disease will progress to CRPC, where the patient is a candidate for future treatment with a biomarker-directed therapy (PARP inhibitor or checkpoint inhibitor).
Who should receive sequential somatic testing with NGS technologies?	3. The panel recommends that sequential somatic testing may be offered when there has been a meaningful change in the patient's status or treatment plan, especially in cases where prior tests were negative or uninformative (eg, insufficient or low tumor content). (Evidence quality: Moderate; Strength of recommendation: Weak)
What are the strengths and weaknesses of primary tumor archival tissue versus fresh metastatic biopsy tissue versus ctDNA testing for somatic testing?	4. Archival tissue samples are preferred in initial testing. ctDNA is preferred when there is no accessible metastatic site to biopsy or for sequential testing. In the setting of minimal disease burden associated with low ctDNA fraction, metastatic biopsy is preferred. (Evidence quality: Low; Strength of recommendation: Weak)
What are the key therapeutic impacts of germline or somatic testing for single-gene genetic variants (eg, *BRCA1*, *BRCA2*)?	5. Patients with pathogenic germline variants or somatic alterations in *BRCA1* and *BRCA2* demonstrate poorer outcomes but are candidates for treatment with PARP inhibitor monotherapy, PARP inhibitor with ARPI combination therapy, and platinum-based agents. (Evidence quality: High; Strength of recommendation: Strong)
What are the key prognostic impacts of germline or somatic testing?	6. Treatment recommendations should not be made based on prognostic only biomarkers. However, they may be considered for directing patients to clinical trials. Germline information may still be important for patient counseling, informing hereditary risk for patients and families. (Evidence quality: High; Strength of recommendation: Strong)

NOTE. The strength of the recommendation is defined as follows: Strong: In recommendations for an intervention, the desirable effects of an intervention outweigh its undesirable effects. In recommendations against an intervention, the undesirable effects of an intervention outweigh its desirable effects. All or almost all informed people would make the recommended choice for or against an intervention. Conditional/weak: In recommendations for an intervention, the desirable effects probably outweigh the undesirable effects, but appreciable uncertainty exists. In recommendations against an intervention, the undesirable effects probably outweigh the desirable effects, but appreciable uncertainty exists. Most informed people would choose the recommended course of action, but a substantial number would not.
Abbreviations: ARPI, androgen receptor pathway inhibitor; CRPC, castration-resistant prostate cancer; CSPC, castration-sensitive prostate cancer; ctDNA, circulating tumor DNA; FDA, US Food and Drug Administration; mCSPC, metastatic castration-sensitive prostate cancer; NGS, next-generation sequencing; PARP, poly(ADP-ribose) polymerase; PSA, prostate-specific antigen.

Downloaded from ascopubs.org by 70.187.168.80 on February 2, 2026 from 070.187.168.080
Copyright © 2026 American Society of Clinical Oncology. All rights reserved.

See Figure 1: Germline and somatic genomic testing for metastatic prostate cancer for a visual representation of these recommendations.

DISCUSSION

As next-generation DNA sequencing technologies are becoming commonplace for patients with various malignancies, it is important to evaluate the evidence and clinical implications for both germline and somatic genomic testing for patients with individual tumor types. This guideline was formulated by a multidisciplinary expert panel to evaluate the evidence surrounding germline and somatic genomic testing for patients with metastatic prostate cancer to provide recommendations for several key clinical aspects:

- Why test—the rationale for germline and somatic genomic testing
- Who to test—which patient categories or disease states
- When to test—at initial diagnosis, and/or with disease progression or potential change in care plan
- What test(s) to use—specifically which specimens and assays to consider, considering how that may impact subsequent recommendations for clinical care.

Although 1,713 papers were found in our PubMed search, only 14 met the stringent criteria for inclusion in this document. The Expert Panel used these data to answer key research questions. The Expert Panel supports germline testing all patients with metastatic prostate cancer, in light of the downstream cascade testing implications and potential for

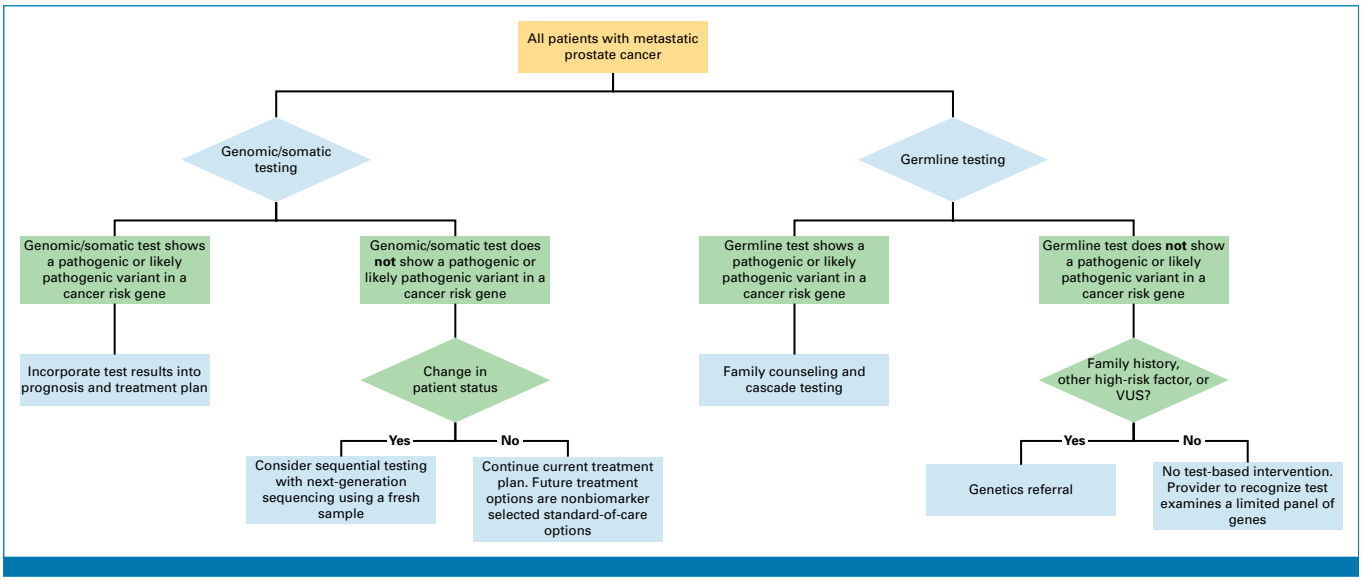

FIG 1. Germline and somatic genomic testing for metastatic prostate cancer. VUS, variants of uncertain significance.

Downloaded from ascopubs.org by 70.187.168.80 on February 2, 2026 from 070.187.168.080
Copyright © 2026 American Society of Clinical Oncology. All rights reserved.

identifying cancer predisposition in family members, recognizing there are available options to reduce or manage such risk. The Expert Panel also supports somatic testing for all patients with metastatic prostate cancer who are eligible for, or who have potential to soon be eligible for, targeted therapy.

Even in cases when germline testing is positive for an actionable pathogenic mutation (eg, *BRCA2*), somatic testing may be useful to confirm loss of heterozygosity and/or identify other actionable somatic alterations. PARP inhibitors have been shown to improve outcomes in patients with mCRPC associated with HRD that can be identified via somatic or germline testing. Although it did not meet criteria for inclusion in this manuscript, the limited data for the use of anti–PD-1 immunotherapy for patients with mismatch repair deficiency, microsatellite instability, and/or high tumor mutational burden (TMB-H) prostate cancers deserve mention, given that germline and somatic testing can also unearth populations who can have meaningful, durable responses to these treatments.

The panel recognizes the data supporting repeat testing are limited. While DNA repair gene mutations tend to be truncal,[18] the data from a recent retrospective analysis from the PROMISE somatic database showed that 11% of patients who underwent repeat testing with either ctDNA or metastatic biopsy had new actionable findings, for example, HRD, mismatch repair deficient (MMRd), microsatellite instability high (MSI-H), or TMB-H.[19] Therefore, when deemed appropriate, retesting in a patient who had prior somatic findings that were either not actionable or were uninformative with ctDNA, a metastatic biopsy was considered low- to moderate-level evidence and thus has a weak recommendation. In the future, with newer agents (eg, CYP11A antagonists, androgen receptor degraders, and N-terminal domain androgen receptor inhibitors) that may address other targets with more evolutionary flux contributing to treatment resistance, the Expert Panel will likely accumulate higher levels of evidence to make a stronger recommendation. The panel also recognized the clinical dilemma of finding variants of uncertain significance (VUS), which may be reclassified as either benign or pathogenic in the future, and which could have clinical and/or familial significance. Finally, this panel emphasizes the importance of distinguishing between a prognostic and a predictive biomarker. Whereas some HRD genes, especially *BRCA2*, have a role as both prognostic and predictive biomarkers, many more genes identified on NGS panel testing have, as yet, uncertain significance or are currently candidates as prognostic biomarkers. For example, patients with metastatic castration-sensitive prostate cancer and *SPOP* gene mutations have better outcomes than patients with wildtype *SPOP*,[20] yet at this time, there is insufficient evidence for *SPOP* alteration status to be used to guide treatment decision making. Alternatively, compound genomic alterations of *TP53*, *PTEN*, and *Rb1* tumor suppressor genes are associated with worse prognosis.[21] This panel feels strongly that treatment recommendations should not be formulated by these prognostic-only biomarkers, but appropriate family counseling and cascade testing recommendations should be considered when clinically impactful findings are discovered in the germline. For patients who carry poor or favorable prognostic results, the development and referral for intensification and deintensification clinical trials, respectively, are recommended.

Limitation of the Research and Future Research

The available data have largely been taken from randomized phase III trials, which are in turn generated from clinical trial patients with relatively few comorbid conditions and may not represent the scope of diversity (clinical, ethnic, and

Downloaded from ascopubs.org by 70.187.168.80 on February 2, 2026 from 070.187.168.080
Copyright © 2026 American Society of Clinical Oncology. All rights reserved.

racial) of clinical practice. Moreover, these data lack the reporting granularity of aspects such as somatic versus germline, zygosity, functionality of alterations, etc. Now that germline genetic testing, genomic testing, and targeted agents are available as standard of care, more data will become available. Some potentially actionable alterations lack randomized phase III data, and thus were not included in this analysis, but warrant mention. Notably, MMRd, MSI-H, and TMB-H carry US Food and Drug Administration (FDA) approval and have been reported to result in significant and durable clinical responses.

It is worth noting that commercial assays may have variable quality and analytic coverage (sensitivity, specificity, ability to evaluate tumor content, variant-allele fractions, and include and/or exclude germline findings), and are also in evolution, for example, bioinformatics pipelines and correction for clonal hematopoiesis of indeterminant potential (CHIP).[22] At present, somatic testing is guided by clinical actionability and/or predictive value. Although virtually all genomic testing platforms provide multigene analysis, there are limited data regarding the prognostic and treatment-predictive significance of co-occurring alterations.

Future research includes implementation science in diverse clinic practice settings. This is relevant as targeted therapies are being studied in earlier disease states (eg, metastatic castration-sensitive prostate cancer, biochemical recurrence, neoadjuvant). More research is needed to determine how to best integrate genomics with newer analytes of interest (eg, immunohistochemistry, RNA-based transcriptomics, methylation, fragmentomics, etc).[23] Some important areas of interest include minimal residual disease and dynamic ctDNA fraction monitoring to identify early disease response and progression. These may have the capability to identify treatment resistance changes in biology, such as neuroendocrine transdifferentiation or reversion mutations.[24] As novel prognostic and predictive biomarkers mature, clinical trial opportunities for studies evaluating intensification, deintensification, and other tailoring of treatments will become more feasible.

PATIENT AND CLINICIAN COMMUNICATION

For recommendations and strategies to optimize patient-clinician communication, see Patient-Clinician Communication: American Society of Clinical Oncology Consensus Guideline.[25] Patient and clinician communication is critical in the context of germline and somatic testing for metastatic prostate cancer. Somatic testing may identify genetic variants that may be of germline origin, and this information should be conveyed to patients prior to ordering somatic testing. Clues that a somatic variant may be of germline origin include high variant allele fraction, pathogenic/likely pathogenic variant (P/LPV) in a cancer susceptibility gene (eg, *BRCA2*), correlating family cancer history, or correlating ancestry. Whether conducting confirmatory germline testing from initial somatic testing or initiating independent germline testing, ASCO, National Comprehensive Cancer Center (NCCN), National Society of Genetic Counselors, and other professional organizations have consistently emphasized the need for patients to have informed consent for germline genetic testing[26-28] based on the hereditary nature of germline testing. While germline testing may be initiated in the context of therapeutic decision making for metastatic prostate cancer, results may identify additional cancer risk for patients and inform blood relatives about their cancer risks. As such, clinicians are recommended to ensure the following key elements are conveyed to patients prior to germline testing: purpose of germline testing, hereditary nature of testing, potential to identify additional cancer risks for the patient and their family, options for panel testing, potential results (P/LPVs, VUS, negative, or inconclusive), the Genetic Information Nondiscrimination Act law, and the importance of cascade testing in families with P/LPV identified.[27,29,30] Given clinician time constraints, several studies have reported use of videos or web-based technologies to deliver pretest information to patients in concert with clinical teams to answer questions and facilitate the informed consent process.[31-34] Patients expressing concerns regarding germline testing, significant anxiety, or who prefer to see a genetic counselor or genetics professional should be referred accordingly. It is strongly recommended that clinicians convey to patients that they will be referred to a genetics professional to discuss cancer risks and cascade testing in families if a P/LPV is identified. There may also be other reasons to refer patients with VUS, depending on the clinician comfort level with discussion of VUS and appropriate clinical management of patients. Finally, it is recommended that clinical teams proactively collaborate with cancer genetics programs to establish and indicate clear patient care pathways for pre- and post-test referral to cancer genetics.

HEALTH EQUITY CONSIDERATIONS

Social determinants of health, defined by the World Health Organization as the conditions in which an individual is born, grows, lives, works, and ages, can undermine ASCO's expert recommendations on best practices for cancer prevention, screening, disease management, and palliative and supportive care.[35,36] It is important to acknowledge many people in the United States and elsewhere receive suboptimal access and quality of cancer care due to the long-term impact of structural factors, societal factors, and the unequal distribution of wealth.[37]

In the United States, many patients remain unable to reap the benefits of innovative prevention and early detection programs, biomarker testing, and new cancer therapies due to structural barriers including lack of transportation, stable housing, and adequate insurance coverage as well as food insecurity, health literacy, proximity to a dedicated cancer center, and cost of treatment and other services.[38] Additionally, people of sexual and gender minority experience stigma along with barriers to cancer screening, prevention,

and treatment that contribute to these cancer disparities.[39] Disparities widen in those who are also from a racial or ethnic minority, underscoring the influence of intersectionality in cancer health disparities.[40]

Further, geographic disparities can also impact the quality of care patients receive. Rural patients with prostate cancer are more likely to experience higher mortality rates compared to nonrural patients.[41] This can be attributed, in part, to a lower density of specialists and dedicated cancer centers, as only 21% of nonmetropolitan counties in the United States have one or more practicing oncologists.[42]

In prostate cancer, people with African ancestry are more likely to be diagnosed with and die from the disease.[43] Although scientific and clinical progress has been made, there is a large unmet need to better understand prostate cancer genomics, epigenomics, and other risk-related variations by race and ethnicity. There are also opportunities to learn from groups such as Asian American Pacific Islander and American Indian and Alaska Native populations who have considerably lower rates of incidence and death from the disease,[43] although it will be important to consider appropriate disaggregation of these heterogeneous populations for both social and biological studies.[44]

While patients of African ancestry have the highest known incidence and mortality among any racial or ethnic group, the greatest absolute reduction in age-adjusted prostate cancer mortality has been achieved for men of African ancestry during the prostate cancer screening era in the United States.[43] Multiple studies indicate that outcomes for men with African ancestry are similar to other groups when there is equal access to high-quality health care.[45-47] These data highlight the critical importance of ambitious and innovative approaches to transform health care systems striving for equity in access, quality of care, and outcomes.[45]

Most comprehensive prostate tumor molecular profiling studies have been conducted on men of European ancestry, such as The Cancer Genome Atlas prostate cancer cohort, which included only 13% African American men.[48] There are other large cohorts, including men of African ancestry, from which considerable insight and progress have been made to identify specific genome-related factors.[49-52]

The frequency of several tumor genomic alterations, including *ERG* rearrangement, *PTEN* deletion, and *SPOP* mutation, differs between Caucasian and African American populations.[53-56] While the overall frequency of germline pathogenic variants do not appear to differ by race in metastatic prostate cancer, *BRCA1* germline mutations and somatic mutations have been observed at higher frequency in African American patients.[57,58] Notably, the initial study demonstrating activity with the PARP inhibitor olaparib in homologous recombination repair deficient prostate cancer did not report race[59] and there were <5% African American patients in the trial investigating the use of the PARP inhibitor, rucaparib.[16] Similarly, the trial demonstrating benefit with the immune checkpoint inhibitor, pembrolizumab, in cancers with microsatellite instability included <10% African American patients.[60] Therefore, it is not clear if responses to PARP or checkpoint inhibition among patients with HRD or MSI-H prostate cancer vary by race. Interestingly, there are data suggesting Black and Hispanic men with nonmetastatic castration-resistant prostate cancer have improved outcomes compared to other racial and ethnic groups in an equal access health care system[61]; similarly, a lower risk of metastases and cancer-specific mortality have been observed in Black men receiving local therapy.[62,63]

The panel strongly encourages ongoing research around the genetic and molecular basis of prostate cancer to improve the management of people with health inequities, while also identifying and implementing large-scale strategies to enhance health care access, coverage, and quality.

MULTIPLE CHRONIC CONDITIONS

Creating evidence-based recommendations to inform treatment of patients with additional chronic conditions, a situation in which the patient may have two or more such conditions—referred to as multiple chronic conditions (MCC)—is challenging. Patients with MCC are a complex and heterogeneous population, making it difficult to account for all possible permutations and develop specific recommendations for care. In addition, the best available evidence for treating index conditions, such as cancer, is often from clinical trials whose study selection criteria may exclude patients with comorbid conditions to avoid potential interactions or confounding effects associated with MCC. As a result, the applicability of outcomes from these studies is limited and constrains the ability of expert groups to make recommendations for care in this heterogeneous patient population.

Since many patients present with MCC, treatment plans need to consider the complexity and uncertainty created by the MCC and highlights the importance of shared decision making regarding guideline use and implementation. Therefore, in consideration of recommended care for the target index condition, clinicians should review all other chronic conditions present in the patient and take those conditions into account when formulating the treatment and follow-up plan.

In patients with prostate cancer and MCC, ctDNA may be favored over metastatic biopsy since the latter may be less safe and/or feasible. In addition to analytic considerations discussed previously such as tumor content, tumor heterogeneity, and CHIP interference, it is also important to recognize that other concurrent conditions, for example, chronic lymphocytic leukemia, or concurrent infections or inflammatory processes may also confound ctDNA results and interpretation.

Downloaded from ascopubs.org by 70.187.168.80 on February 2, 2026 from 070.187.168.080
Copyright © 2026 American Society of Clinical Oncology. All rights reserved.

In light of these considerations, practice guidelines should provide information on how to apply the recommendations for patients with MCC as a qualifying statement for recommended care. This may mean that some or all of the recommended care options are modified or not applied, as determined by best practice in consideration of any MCC. See the Data Supplement: Multiple Chronic Conditions for the most common chronic conditions.

COST IMPLICATIONS

Increasingly, individuals with cancer are required to pay a larger proportion of their treatment costs through deductibles and co-insurance.[64,65] Higher patient out-of-pocket costs have been shown to be a barrier to initiating and adhering to recommended cancer treatments.[66,67]

Discussion of cost can be an important part of shared decision making.[68] Clinicians should discuss with patients the use of less expensive alternatives when it is practical and feasible for treatment of the patient's disease and there are two or more treatment options that are comparable in terms of benefits and harms.[68]

The integration of germline and somatic tests into routine clinical practice has significant cost implications, both in terms of direct testing costs and broader health care expenditures. The testing methodologies range from targeted gene panels, whole-exome sequencing (WES), to whole-genome sequencing (WGS), which have variable cost implications.

Common Types of Germline Testing and Costs

- Targeted Gene Panels: These panels focus on a specific set of genes known to be associated with prostate cancer. The cost of targeted gene panels generally ranges from $250 in US dollars (USD) to $3,000 USD depending on the number of genes tested and the complexity of the assay. This is the most common method used for germline testing. Targeted gene panels are the current standard of care.
- WES: WES examines the entire coding region of the genome, capturing all possible mutations in exons. This method is more comprehensive than targeted panels but comes at a higher cost, typically between $1,000 USD and $3,000 USD. WES, along with WGS, is one of the two standards of care for pediatric patients.
- WGS: WGS provides a complete picture of the patient's genetic makeup, including both coding and noncoding regions. WGS is the most expensive option, with costs ranging from $3,000 USD to $5,000 USD. WGS, along with WES, is one of the two standards of care for pediatric patients.

Common Types of Somatic Testing and Costs

- NGS Panels: NGS panels designed for somatic testing typically assess a broad range of genes that are relevant to cancer biology. The cost for these panels generally ranges from $1,000 USD to $4,000 USD depending on the extent of the panel and the number of genes included.
- Liquid biopsy: A less invasive method, liquid biopsy involves the analysis of ctDNA from blood samples. While convenient, the costs of liquid biopsy are similar to those of traditional biopsy methods, ranging from $1,000 USD to $3,000 USD per test.
- Comprehensive genomic profiling (CGP): CGP provides a complete view of the tumor's genomic landscape, assessing mutations, copy-number alterations, and gene rearrangements. The cost for CGP is typically higher, often exceeding $4,000 USD per test.

The cost estimates of these tests are estimates with broad ranges due to differences in technology and use of a commercial platform or if an in-house solution, if available. The costs for genetic testing are likely to decrease over time. The occurrences of (somatic) testing may also contribute to cost, as repeat biopsies may be used to monitor the evolution of the tumor genome over time. Furthermore, the cost estimates provided here do not reflect out-of-pocket costs, which may vary widely as tests billed through insurance are influenced by deductibles, copays, or percentages of services rendered. When discussing financial issues and concerns, patients should be made aware of financial counseling services available to address these complexities.[68]

In summary, the financial implications of germline and somatic testing for patients with metastatic prostate cancer are multifaceted, involving both direct testing costs and broader economic considerations. While these tests are associated with upfront costs, they offer the potential for cost-savings by enabling personalized treatment strategies that improve outcomes and reduce unnecessary interventions. Continued research into the cost-effectiveness of these testing modalities is essential to optimize their use in clinical practice and ensure that all patients have access to the benefits of precision oncology.

GUIDELINE IMPLEMENTATION

ASCO guidelines are developed for implementation across health settings. Each ASCO guideline includes a community oncology representation on the panel. The additional role of this community oncologist on the guideline panel is to assess the suitability of the recommendations to implementation in the community setting, but also to identify any other barrier to implementation a reader should be aware of. Barriers to implementation include the need to increase awareness of the guideline recommendations among frontline practitioners and survivors of cancer and caregivers, and also to provide adequate services in resource-limited settings. The guideline recommendations table and accompanying tools (available at www.asco.org/genitourinary-cancer-guidelines) were designed to facilitate

Downloaded from ascopubs.org by 70.187.168.80 on February 2, 2026 from 070.187.168.080
Copyright © 2026 American Society of Clinical Oncology. All rights reserved.

implementation of recommendations. This guideline will be distributed widely through ASCO State Societies, to reach oncologists and other clinicians in community practice. ASCO guidelines are posted on the ASCO website and most often published in the *Journal of Clinical Oncology*.

Based on the availability of drugs targeting actionable alterations in patients with metastatic prostate cancer, the panel strongly recommends genomic profiling through tumor tissue or ctDNA. However, a real-world nationwide study showed that only 29.3% of patients with metastatic prostate cancer received NGS between 2015 and 2022, with testing rates increasing from 19% to 27.1% during this period. Black and Hispanic patients, those with low socioeconomic status, and those covered by Medicaid, Medicare, or other government insurance programs were significantly less likely to undergo sequencing.[69] Other factors associated with not receiving NGS testing include patients receiving care in community practice settings, patients older than 65 years, and patients with poor performance status.[70] Integrating NGS testing into the standard-of-care management for patients with metastatic prostate cancer and improving equitable access to health care services remain crucial unmet needs.

ASCO believes that cancer clinical trials are vital to inform medical decisions and improve cancer care, and that all patients should have the opportunity to participate.

ADDITIONAL RESOURCES

For current information, including selected updates, supplements, slide sets, and clinical tools and resources, visit http://www.asco.org/genitourinary-cancer-guidelines. The Data Supplement for this guideline includes additional evidence tables, details of the included trials, and all other outcomes. Guideline recommendations and algorithms are also available in the free ASCO Guidelines app (available for download in the Apple App Store and Google Play Store). Listen to key recommendations and insights from panel members on the ASCO Guidelines podcast. The Methodology Manual (available at www.asco.org/guideline-methodology) provides additional information about the methods used to develop this guideline. Patient information is available at www.cancer.org.

ASCO welcomes your comments on this guideline, including implementation challenges, new evidence, and how this guideline impacts you. To provide feedback, contact us at guidelines@asco.org. Comments may be incorporated into a future guideline update. To submit new evidence or suggest a topic for guideline development, complete the form available at www.asco.org/guidelines.

RELATED ASCO GUIDELINES

- Palliative Care for Patients with Cancer[71] (http://ascopubs.org/doi/10.1200/JCO.24.00542)
- Patient-Clinician Communication[25] (http://ascopubs.org/doi/10.1200/JCO.2017.75.2311)
- Molecular Biomarkers in Localized Prostate Cancer[72] (https://ascopubs.org/doi/10.1200/JCO.19.02768)
- Selection of Germline Genetic Testing Panels in Patients with Cancer[73] (https://ascopubs.org/doi/10.1200/JCO.24.00662)
- Management of Hereditary Breast Cancer[74] (https://ascopubs.org/doi/10.1200/JCO.20.00299)
- Germline Testing in Patients with Breast Cancer[75] (https://ascopubs.org/doi/10.1200/JCO.23.02225)
- Somatic Genomic Testing in Patients with Metastatic or Advanced Cancer[76] (https://ascopubs.org/doi/10.1200/JCO.21.02767)
- Germline and Somatic Tumor Testing in Epithelial Ovarian Cancer[77] (https://ascopubs.org/doi/10.1200/JCO.19.02960)

GENDER-INCLUSIVE LANGUAGE

ASCO is committed to promoting the health and well-being of individuals regardless of sexual orientation or gender identity. Transgender and nonbinary people, in particular, may face barriers to care. One way exclusiveness or lack of accessibility may be communicated is through gendered language that makes presumptive links between gender and anatomy.[78-81] With the acknowledgment that ASCO guidelines may impact the language used in clinical and research settings, ASCO is committed to creating gender-inclusive guidelines. For this reason, guideline authors use gender-inclusive language whenever possible throughout the guidelines. In instances in which the guideline draws upon data based on gendered research (eg, studies regarding men with prostate cancer), the guideline authors describe the characteristics and results of the research as reported.

This is especially important in prostate cancer, as prior reports suggest it is the second most common neoplasm among sexual minorities, with one in six gay and bisexual men diagnosed with prostate cancer in their lifetime. Furthermore, transgender individuals are less likely to undergo prostate cancer screening and make informed decisions about their health care. Additionally, sexual minorities experience poorer quality-of-life outcomes following prostate cancer treatment. Therefore, it is crucial to mitigate these disparities and improve access to genomic profiling among sexual minorities to provide them with a better chance of benefiting from targeted therapies.

Downloaded from ascopubs.org by 70.187.168.80 on February 2, 2026 from 070.187.168.080
Copyright © 2026 American Society of Clinical Oncology. All rights reserved.

AFFILIATIONS

[1]University of Washington and Fred Hutchinson Cancer Center, Seattle, WA
[2]American Society of Clinical Oncology, Alexandria, VA
[3]Huntsman Cancer Institute at the University of Utah, Salt Lake City, UT
[4]University of Chicago, Chicago, IL
[5]Duke Cancer Institute, Durham, NC
[6]Dana Farber Cancer Institute, Boston, MA
[7]Yale School of Medicine and Yale Cancer Center, New Haven, CT
[8]UH Cleveland Medical Center, Cleveland, OH
[9]UM/Sylvester Comprehensive Cancer Center, Miami, FL
[10]Kaohsiung Medical University Hospital and Kaohsiung Medical University, Kaohsiung City, Taiwan
[11]Patient Representative, Chapter President UsTOO Las Vegas, Prostate Cancer Education and Support, Las Vegas, NV
[12]Jefferson Einstein Medical Center, Sidney Kimmel Cancer Center of Thomas Jefferson University, Philadelphia, PA

CORRESPONDING AUTHOR

American Society of Clinical Oncology; e-mail: guidelines@asco.org.

EDITOR'S NOTE

This ASCO Clinical Practice Guideline provides recommendations, with comprehensive review and analyses of the relevant literature for each recommendation. Additional information, including a supplement with additional evidence tables, slide sets, clinical tools and resources, and links to patient information at www.cancer.org, is available at www.asco.org/genitourinary-cancer-guidelines.

EQUAL CONTRIBUTION

E.Y.Y. and E.J.T. were Expert Panel co-chairs.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Disclosures provided by the authors are available with this article at DOI https://doi.org/10.1200/JCO-24-02608.

AUTHOR CONTRIBUTIONS

Conception and design: All authors
Administrative support: R. Bryan Rumble
Collection and assembly of data: All authors
Data analysis and interpretation: All authors
Manuscript writing: All authors
Final approval of manuscript: All authors
Accountable for all aspects of the work: All authors

ACKNOWLEDGMENT

The Expert Panel wishes to thank the two EBMC reviewers, Dr Shanthi Sivendran, MD, MSCR, MBA, and Dr Himanshu Nagar, MD, MS, and the Evidence Based Medicine Committee for their thoughtful reviews and insightful comments on this guideline. The Expert Panel also wishes to thank Mark Somerfield, PhD, and Thomas K. Oliver for providing comments on an earlier draft of the manuscript.

Downloaded from ascopubs.org by 70.187.168.80 on February 2, 2026 from 070.187.168.080
Copyright © 2026 American Society of Clinical Oncology. All rights reserved.

REFERENCES

1. Siegel RL, Giaquinto AN, Jemal A: Cancer statistics, 2024. CA Cancer J Clin 74:12-49, 2024
2. Balshem H, Helfand M, Schünemann HJ, et al: GRADE guidelines: 3. Rating the quality of evidence. J Clin Epidemiol 64:401-406, 2011
3. Higgins JPT, López-López JA, Becker BJ, et al: Synthesising quantitative evidence in systematic reviews of complex health interventions. BMJ Glob Health 4:e000858, 2019
4. de Bono J, Mateo J, Fizazi K, et al: Olaparib for metastatic castration-resistant prostate cancer. N Engl J Med 382:2091-2102, 2020
5. Antonarakis ES, Gomella LG, Petrylak DP: When and how to use PARP inhibitors in prostate cancer: A systematic review of the literature with an update on on-going trials. Eur Urol Oncol 3:594-611, 2020
6. Casanova-Salas I, Athie A, Boutros PC, et al: Quantitative and qualitative analysis of blood-based liquid biopsies to inform clinical decision-making in prostate cancer. Eur Urol 79:762-771, 2021
7. Gleicher S, Kauffman EC, Kotula L, et al: Implications of high rates of metastatic prostate cancer in BRCA2 mutation carriers. Prostate 76:1135-1145, 2016
8. Lang SH, Swift SL, White H, et al: A systematic review of the prevalence of DNA damage response gene mutations in prostate cancer. Int J Oncol 55:597-616, 2019
9. Loeb S, Giri VN: Clinical implications of germline testing in newly diagnosed prostate cancer. Eur Urol Oncol 4:1-9, 2021
10. Marino F, Totaro A, Gandi C, et al: Germline mutations in prostate cancer: A systematic review of the evidence for personalized medicine. Prostate Cancer Prostatic Dis 26:655-664, 2023
11. Shore N, Oliver L, Shui I, et al: Systematic literature review of the epidemiology of advanced prostate cancer and associated homologous recombination repair gene alterations. J Urol 205:977-986, 2021
12. Van der Eecken K, Vanwelkenhuyzen J, Deek MP, et al: Tissue- and blood-derived genomic biomarkers for metastatic hormone-sensitive prostate cancer: A systematic review. Eur Urol Oncol 4: 914-923, 2021
13. Agarwal N, Azad AA, Carles J, et al: Talazoparib plus enzalutamide in men with first-line metastatic castration-resistant prostate cancer (TALAPRO-2): A randomised, placebo-controlled, phase 3 trial. Lancet 402:291-303, 2023
14. Chi KN, Rathkopf D, Smith MR, et al: Niraparib and abiraterone acetate for metastatic castration-resistant prostate cancer. J Clin Oncol 41:3339-3351, 2023
15. Clarke NW, Armstrong AJ, Thiery-Vuillemin A, et al: Abiraterone and olaparib for metastatic castration-resistant prostate cancer. NEJM Evid 1:EVIDoa2200043, 2022
16. Fizazi K, Piulats JM, Reaume MN, et al: Rucaparib or physician's choice in metastatic prostate cancer. N Engl J Med 388:719-732, 2023
17. Hussain M, Kocherginsky M, Agarwal N, et al: Abiraterone, olaparib, or abiraterone + olaparib in first-line metastatic castration-resistant prostate cancer with DNA repair defects (BRCAAway). Clin Cancer Res 30:4318-4328, 2024
18. Schweizer MT, Sivakumar S, Tukachinsky H, et al: Concordance of DNA repair gene mutations in paired primary prostate cancer samples and metastatic tissue or cell-free DNA. JAMA Oncol 7: 1378-1382, 2021
19. Park JJ, Chu A, Li J, et al: Repeat next-generation sequencing testing on progression in men with metastatic prostate cancer can identify new actionable alterations. JCO Precis Oncol 10.1200/PO.23.00567
20. Swami U, Isaacsson Velho P, Nussenzveig R, et al: Association of SPOP mutations with outcomes in men with de novo metastatic castration-sensitive prostate cancer. Eur Urol 78:652-656, 2020
21. Hamid AA, Gray KP, Shaw G, et al: Compound genomic alterations of TP53, PTEN, and RB1 tumor suppressors in localized and metastatic prostate cancer. Eur Urol 76:89-97, 2019
22. Jensen K, Konnick EQ, Schweizer MT, et al: Association of clonal hematopoiesis in DNA repair genes with prostate cancer plasma cell-free DNA testing interference. JAMA Oncol 7:107-110, 2021
23. Herberts C, Wyatt AW, Nguyen PL, et al: Genetic and genomic testing for prostate cancer: Beyond DNA repair. Am Soc Clin Oncol Educ Book 43:e390384, 2023
24. Fonseca NM, Maurice-Dror C, Herberts C, et al: Prediction of plasma ctDNA fraction and prognostic implications of liquid biopsy in advanced prostate cancer. Nat Commun 15:1828, 2024
25. Gilligan T, Coyle N, Frankel RM, et al: Patient-clinician communication: American Society of Clinical Oncology consensus guideline. J Clin Oncol 35:3618-3632, 2017
26. Lu KH, Wood ME, Daniels M, et al: American Society of Clinical Oncology expert statement: Collection and use of a cancer family history for oncology providers. J Clin Oncol 32:833-840, 2014
27. Daly MB, Pal T, Maxwell KN, et al: NCCN guidelines insights: Genetic/familial high-risk assessment: Breast, ovarian, and pancreatic, version 2.2024. J Natl Compr Canc Netw 21:1000-1010, 2023
28. Riley BD, Culver JO, Skrzynia C, et al: Essential elements of genetic cancer risk assessment, counseling, and testing: Updated recommendations of the National Society of Genetic Counselors. J Genet Couns 21:151-161, 2012
29. Giri VN, Knudsen KE, Kelly WK, et al: Implementation of germline testing for prostate cancer: Philadelphia Prostate Cancer Consensus Conference 2019. J Clin Oncol 38:2798-2811, 2020
30. Cheng HH, Sokolova AO, Schaeffer EM, et al: Germline and somatic mutations in prostate cancer for the clinician. J Natl Compr Cancer Netw 17:515-521, 2019

31. Loeb S, Keith SW, Cheng HH, et al: TARGET: A randomized, noninferiority trial of a pretest, patient-driven genetic education webtool versus genetic counseling for prostate cancer germline testing. JCO Precis Oncol 10.1200/PO.23.00552
32. Cheng HH, Sokolova AO, Gulati R, et al: Internet-based germline genetic testing for men with metastatic prostate cancer. JCO Precis Oncol 10.1200/PO.22.00104
33. Russo J, McDougall C, Bowler N, et al: Pretest genetic education video versus genetic counseling for men considering prostate cancer germline testing: A patient-choice study to address urgent practice needs. JCO Precis Oncol 10.1200/PO.21.00238
34. Rana HQ, Stopfer JE, Weitz M, et al: Pretest video education versus genetic counseling for patients with prostate cancer: ProGen, a multisite randomized controlled trial. JCO Oncol Pract 19: 1069-1079, 2023
35. The World Health Organization: Social determinants of health, 2024. https://www.who.int/health-topics/social-determinants-of-health#tab=tab_1
36. Tucker-Seeley R, Abu-Khalaf M, Bona K, et al: Social determinants of health and cancer care: An ASCO policy statement. JCO Oncol Pract 20:621-630, 2024
37. Carlos RC, Obeng-Gyasi S, Cole SW, et al: Linking structural racism and discrimination and breast cancer outcomes: A social genomics approach. J Clin Oncol 40:1407-1413, 2022
38. Hendren S, Chin N, Fisher S, et al: Patients' barriers to receipt of cancer care, and factors associated with needing more assistance from a patient navigator. J Natl Med Assoc 103:701-710, 2011
39. Alpert AB, Scout N, Schabath MB, et al: Gender-and sexual orientation–based inequities: Promoting inclusion, visibility, and data accuracy in oncology. Am Soc Clin Oncol Educ Book 42:542-558, 2022
40. Williams PA, Zaidi SK, Sengupta R: AACR cancer disparities progress report 2022. Cancer Epidemiol Biomarkers Prev 31:1249-1250, 2022
41. Maganty A, Sabik LM, Sun Z, et al: Under treatment of prostate cancer in rural locations. J Urol 203:108-114, 2020
42. Levit LA, Byatt L, Lyss AP, et al: Closing the rural cancer care gap: Three institutional approaches. JCO Oncol Pract 16:422-430, 2020
43. Pack R: Descriptive epidemiology of genitourinary cancers. Semin Oncol Nurs:218-223, 1993
44. Lee RJ, Madan RA, Kim J, et al: Disparities in cancer care and the Asian American population. Oncologist 26:453-460, 2021
45. Nyame YA, Cooperberg MR, Cumberbatch MG, et al: Deconstructing, addressing, and eliminating racial and ethnic inequities in prostate cancer care. Eur Urol 82:341-351, 2022
46. Dess RT, Hartman HE, Mahal BA, et al: Association of Black race with prostate cancer-specific and other-cause mortality. JAMA Oncol 5:975-983, 2019
47. Albers P, Bashir S, Mookerji N, et al: Prostate cancer among Black men in Canada. JAMA Netw Open 7:e2418475, 2024
48. Abeshouse A, Ahn J, Akbani R, et al: The molecular taxonomy of primary prostate cancer. Cell 163:1011-1025, 2015
49. Hoffmann TJ, Graff RE, Madduri RK, et al: Genome-wide association study of prostate-specific antigen levels in 392,522 men identifies new loci and improves cross-ancestry prediction. medRxiv 10.27.23297676
50. Darst BF, Shen J, Madduri RK, et al: Evaluating approaches for constructing polygenic risk scores for prostate cancer in men of African and European ancestry. Am J Hum Genet 110:1200-1206, 2023
51. Darst BF, Hughley R, Pfennig A, et al: A rare germline HOXB13 variant contributes to risk of prostate cancer in men of African ancestry. Eur Urol 81:458-462, 2022
52. Matejcic M, Patel Y, Lilyquist J, et al: Pathogenic variants in cancer predisposition genes and prostate cancer risk in men of African ancestry. JCO Precis Oncol 4:32-43, 2020
53. Khani F, Mosquera JM, Park K, et al: Evidence for molecular differences in prostate cancer between African American and Caucasian men. Clin Cancer Res 20:4925-4934, 2014
54. Petrovics G, Li H, Stümpel T, et al: A novel genomic alteration of LSAMP associates with aggressive prostate cancer in African American men. EBioMedicine 2:1957-1964, 2015
55. Lindquist KJ, Paris PL, Hoffmann TJ, et al: Mutational landscape of aggressive prostate tumors in African American men. Cancer Res 76:1860-1868, 2016
56. Yuan J, Kensler KH, Hu Z, et al: Integrative comparison of the genomic and transcriptomic landscape between prostate cancer patients of predominantly African or European genetic ancestry. PLoS Genet 16:e1008641, 2020
57. Ledet EM, Burgess EF, Sokolova AO, et al: Comparison of germline mutations in African American and Caucasian men with metastatic prostate cancer. Prostate 81:433-439, 2021
58. Mahal BA, Alshalalfa M, Kensler KH, et al: Racial differences in genomic profiling of prostate cancer. N Engl J Med 383:1083-1085, 2020
59. Mateo J, Carreira S, Sandhu S, et al: DNA-repair defects and olaparib in metastatic prostate cancer. N Engl J Med 373:1697-1708, 2015
60. Le DT, Uram JN, Wang H, et al: PD-1 blockade in tumors with mismatch-repair deficiency. N Engl J Med 372:2509-2520, 2015
61. Rasmussen KM, Patil V, Li C, et al: Survival outcomes by race and ethnicity in veterans with nonmetastatic castration-resistant prostate cancer. JAMA Netw Open 6:e2337272, 2023
62. Yamoah K, Lee KM, Awasthi S, et al: Racial and ethnic disparities in prostate cancer outcomes in the veterans affairs health care system. JAMA Netw Open 5:e2144027, 2022
63. McKay RR, Sarkar RR, Kumar A, et al: Outcomes of Black men with prostate cancer treated with radiation therapy in the Veterans Health Administration. Cancer 127:403-411, 2021
64. Schnipper LE, Davidson NE, Wollins DS, et al: Updating the American Society of Clinical Oncology value framework: Revisions and reflections in response to comments received. J Clin Oncol 34: 2925-2934, 2016
65. Schnipper LE, Davidson NE, Wollins DS, et al: American Society of Clinical Oncology statement: A conceptual framework to assess the value of cancer treatment options. J Clin Oncol 33: 2563-2577, 2015
66. Dusetzina SB, Winn AN, Abel GA, et al: Cost sharing and adherence to tyrosine kinase inhibitors for patients with chronic myeloid leukemia. J Clin Oncol 32:306-311, 2014
67. Streeter SB, Schwartzberg L, Husain N, et al: Patient and plan characteristics affecting abandonment of oral oncolytic prescriptions. J Oncol Pract 7:46s-51s, 2011
68. Meropol NJ, Schrag D, Smith TJ, et al: American Society of Clinical Oncology guidance statement: The cost of cancer care. J Clin Oncol 27:3868-3874, 2009
69. Chehade CH, Jo Y, Gebrael G, et al: Trends and disparities in next-generation sequencing in metastatic prostate and urothelial cancers. JAMA Netw Open 7:e2423186, 2024
70. Barata PC, Assayag J, Li B, et al: Genetic testing in men with metastatic castration-resistant prostate cancer. JAMA Oncol 10:975-977, 2024
71. Sanders JJ, Temin S, Ghoshal A, et al: Palliative care for patients with cancer: ASCO guideline update. J Clin Oncol 42:2336-2357, 2024
72. Eggener SE, Rumble RB, Armstrong AJ, et al: Molecular biomarkers in localized prostate cancer: ASCO guideline. J Clin Oncol 38:1474-1494, 2020
73. Tung N, Ricker C, Messersmith H, et al: Selection of germline genetic testing panels in patients with cancer: ASCO guideline. J Clin Oncol 42:2599-2615, 2024
74. Tung NM, Boughey JC, Pierce LJ, et al: Management of hereditary breast cancer: American society of clinical oncology, American society for radiation oncology, and society of surgical oncology guideline. J Clin Oncol 38:2080-2106, 2020
75. Bedrosian I, Somerfield MR, Achatz MI, et al: Germline testing in patients with breast cancer: ASCO–Society of Surgical Oncology guideline. J Clin Oncol 42:584-604, 2024
76. Chakravarty D, Johnson A, Sklar J, et al: Somatic genomic testing in patients with metastatic or advanced cancer: ASCO provisional clinical opinion. J Clin Oncol 40:1231-1258, 2022
77. Konstantinopoulos PA, Norquist B, Lacchetti C, et al: Germline and somatic tumor testing in epithelial ovarian cancer: ASCO guideline. J Clin Oncol 38:1222-1245, 2020
78. Alpert AB, Manzano C, Ruddick R: Degendering oncologic care and other recommendations to eliminate barriers to care for transgender people with cancer. ASCO Daily News. Jan 19:2552-2558, 2021
79. Alpert AB, Gampa V, Lytle MC, et al: I'm not putting on that floral gown: Enforcement and resistance of gender expectations for transgender people with cancer. Patient Educ Couns 104:2552-2558, 2021
80. UCSF Gender Affirming Health Program: Transgender Care & Treatment Guidelines. Terminology & Definitions. San Francisco, CA, University of California, 2016
81. Advocates for Trans Equality: Understanding transgender people: The basics, 2023. https://transequality.org/issues/resources/understanding-transgender-people-the-basics
82. Schünemann H, Brożek J, Guyatt G, Oxman A (eds): GRADE Handbook, 2013. https://gdt.gradepro.org/app/handbook/handbook.html#h.svwngs6pm0f2

Downloaded from ascopubs.org by 70.187.168.80 on February 2, 2026 from 070.187.168.080
Copyright © 2026 American Society of Clinical Oncology. All rights reserved.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Germline and Somatic Genomic Testing for Metastatic Prostate Cancer: ASCO Guideline

The following represents disclosure information provided by authors of this manuscript. All relationships are considered compensated unless otherwise noted. Relationships are self-held unless noted. I = Immediate Family Member, Inst = My Institution. Relationships may not relate to the subject matter of this manuscript. For more information about ASCO's conflict of interest policy, please refer to www.asco.org/rwc or ascopubs.org/jco/authors/author-center.

Open Payments is a public database containing information reported by companies about payments made to US-licensed physicians (Open Payments).

Evan Y. Yu
Consulting or Advisory Role: Janssen, Bayer, Merck, Advanced Accelerator Applications, Oncternal Therapeutics, Bristol Myers Squibb, Loxo/Lilly
Research Funding: Dendreon (Inst), Merck (Inst), Seagen (Inst), Blue Earth Diagnostics (Inst), Bayer (Inst), Lantheus Medical Imaging (Inst), Tyra Biosciences (Inst), Oncternal Therapeutics (Inst)

Neeraj Agarwal
Research Funding: Bayer (Inst), Bristol Myers Squibb (Inst), Takeda (Inst), Pfizer (Inst), Exelixis (Inst), Amgen (Inst), AstraZeneca (Inst), Calithera Biosciences (Inst), Celldex (Inst), Eisai (Inst), Genentech (Inst), Immunomedics (Inst), Janssen (Inst), Merck (Inst), Lilly (Inst), Nektar (Inst), ORIC Pharmaceuticals (Inst), ORIC Pharmaceuticals (Inst), crispr therapeutics (Inst), Arvinas (Inst), Gilead Sciences (Inst)
Travel, Accommodations, Expenses: Pfizer, Exelixis

Heather H. Cheng
Consulting or Advisory Role: Janssen Oncology
Research Funding: Sanofi (Inst), Janssen (Inst), Clovis Oncology (Inst), Color Genomics Foundation (Inst), Medivation/Astellas (Inst), Phosplatin Therapeutics (Inst)
Patents, Royalties, Other Intellectual Property: UpToDate
Open Payments Link: https://openpaymentsdata.cms.gov/physician/746939

Scott E. Eggener
Consulting or Advisory Role: Candel Therapeutics, A3P Biomedical, Cellvax, MetasTx, OptumHealth, Janssen, Lantheus Medical Imaging
Research Funding: Candel Therapeutics (Inst), Janssen (Inst)
Open Payments Link: https://openpaymentsdata.cms.gov/physician/492941

Himisha Beltran
Consulting or Advisory Role: AstraZeneca, Pfizer, Daicchi Sankyo, Bayer, Sanofi, Merck, Novartis
Research Funding: Bristol Myers Squibb Foundation (Inst), Daiichi Sankyo (Inst), Novartis (Inst), Circle Pharma (Inst)
Travel, Accommodations, Expenses: Janssen Oncology

Veda N. Giri
Stock and Other Ownership Interests: Novopyxis

Daniel Spratt
Consulting or Advisory Role: Janssen Oncology, AstraZeneca, Boston Scientific, Bayer, Blue Earth Diagnostics, Pfizer, Astellas Scientific and Medical Affairs Inc, Novartis
Research Funding: Janssen (Inst)
Uncompensated Relationships: NCCN
Open Payments Link: https://openpaymentsdata.cms.gov/physician/869226

Brandon Mahal
Honoraria: Foundation Medicine
Other Relationship: Prostate Cancer Foundation, Department of Defense-Prostate Cancer Research Program, American Society for Radiation Oncology

Tony Crispino
Consulting or Advisory Role: Pfizer

Edouard J. Trabulsi
Consulting or Advisory Role: CG Oncology, Johnson & Johnson/Janssen
Speakers' Bureau: Veracyte

No other potential conflicts of interest were reported.

Downloaded from ascopubs.org by 70.187.168.80 on February 2, 2026 from 070.187.168.080
Copyright © 2026 American Society of Clinical Oncology. All rights reserved.

APPENDIX 1. GUIDELINE DISCLAIMER

The Clinical Practice Guidelines and other guidance published herein are provided by ASCO Inc to assist clinicians in clinical decision making. The information herein should not be relied upon as being complete or accurate, nor should it be considered as inclusive of all proper treatments or methods of care or as a statement of the standard of care. With the rapid development of scientific knowledge, new evidence may emerge between the time information is developed and when it is published or read. The information is not continually updated and may not reflect the most recent evidence. The information addresses only the topics specifically identified therein and is not applicable to other interventions, diseases, or stages of diseases. This information does not mandate any particular course of medical care. Further, the information is not intended to substitute for the independent professional judgment of the treating clinician, as the information does not account for individual variation among patients. Recommendations specify the level of confidence that the recommendation reflects the net effect of a given course of action. The use of words like "must," "must not," "should," and "should not" indicates that a course of action is recommended or not recommended for either most or many patients, but there is latitude for the treating physician to select other courses of action in individual cases. In all cases, the selected course of action should be considered by the treating clinician in the context of treating the individual patient. Use of the information is voluntary. ASCO does not endorse third party drugs, devices, services, or therapies used to diagnose, treat, monitor, manage, or alleviate health conditions. Any use of a brand or trade name is for identification purposes only. ASCO provides this information on an "as is" basis and makes no warranty, express or implied, regarding the information. ASCO specifically disclaims any warranties of merchantability or fitness for a particular use or purpose. ASCO assumes no responsibility for any injury or damage to persons or property arising out of or related to any use of this information, or for any errors or omissions.

APPENDIX 2. GUIDELINE AND CONFLICTS OF INTEREST

The Expert Panel was assembled in accordance with ASCO's Conflict of Interest Policy Implementation for Clinical Practice Guidelines ("Policy," found at http://www.asco.org/guideline-methodology). All members of the Expert Panel completed ASCO's disclosure form, which requires disclosure of financial and other interests, including relationships with commercial entities that are reasonably likely to experience direct regulatory or commercial impact as a result of promulgation of the guideline. Categories for disclosure include employment; leadership; stock or other ownership; honoraria, consulting or advisory role; speaker's bureau; research funding; patents, royalties, other intellectual property; expert testimony; travel, accommodations, expenses; and other relationships. In accordance with the Policy, the majority of the members of the Expert Panel did not disclose any relationships constituting a conflict under the Policy.

TABLE A1. Germline and Somatic Genomic Testing for Metastatic Prostate Cancer Guideline Expert Panel Membership

Name	Institution	Area of Expertise
Edouard J. Trabulsi, MD, co-chair	Jefferson Einstein Medical Center, Sidney Kimmel Cancer Center of Thomas Jefferson University, Philadelphia, PA	Surgical Oncology
Evan Y. Yu, MD, co-chair	University of Washington and Fred Hutchinson Cancer Center, Seattle, WA	Medical Oncology
Neeraj Agarwal, MD	Huntsman Cancer Institute at the University of Utah, Salt Lake City, UT	Community Oncology
Himisha Beltran, MD	Dana Farber Cancer Institute, Boston, MA	Medical Oncology
Rhonda L. Bitting, MD	Duke Cancer Institute, Durham, NC	Medical Oncology
Heather H. Cheng, MD, PhD	University of Washington and Fred Hutchinson Cancer Center, Seattle, WA	Medical Oncology
Tony Crispino	Chapter President UsTOO Las Vegas, Prostate Cancer Education and Support, Las Vegas, NV	Patient Representative
Scott E. Eggener, MD	University of Chicago, Chicago, IL	Urology
Veda N. Giri, MD	Yale School of Medicine and Yale Cancer Center, New Haven, CT	Medical Oncology Clinical Cancer Genetics
Kevin Lu, MD	Kaohsiung Medical University Hospital and Kaohsiung Medical University, Kaohsiung City, Taiwan	Urology
Brandon Mahal, MD	UM/Sylvester Comprehensive Cancer Center, Miami, FL	Radiation Oncology
Daniel Spratt, MD	UH Cleveland Medical Center, Cleveland, OH	Radiation Oncology
R. Bryan Rumble, MSc	American Society of Clinical Oncology, Alexandria, VA	ASCO Practice Guideline Staff (Health Research Methods)

Downloaded from ascopubs.org by 70.187.168.80 on February 2, 2026 from 070.187.168.080
Copyright © 2026 American Society of Clinical Oncology. All rights reserved.

© 2025 by American Society of Clinical Oncology

TABLE A2. Evidence Quality and Recommendation Rating Definitions

Term	Definitions
Evidence quality	
High	We are very confident that the true effect lies close to that of the estimate of the effect
Moderate	We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different
Low	Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect
Very low	We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect
Strength of recommendation	
Strong	In recommendations for an intervention, the desirable effects of an intervention outweigh its undesirable effects
	In recommendations against an intervention, the undesirable effects of an intervention outweigh its desirable effects
	All or almost all informed people would make the recommended choice for or against an intervention
Conditional/weak	In recommendations for an intervention, the desirable effects probably outweigh the undesirable effects, but appreciable uncertainty exists
	In recommendations against an intervention, the undesirable effects probably outweigh the desirable effects, but appreciable uncertainty exists
	Most informed people would choose the recommended course of action, but a substantial number would not

NOTE. GRADE Handbook, Schünemann et al.[82]

Downloaded from ascopubs.org by 70.187.168.80 on February 2, 2026 from 070.187.168.080
Copyright © 2026 American Society of Clinical Oncology. All rights reserved.